2013

# COMPUTER SCIENCE AND ENGINEERING

## Paper – CSE – 103

## (Advanced Database Management Systems)

**Full Marks – 100**

*The figures in the margin indicate full marks*

*Candidates are required to give their answers in their own words as far as practicable*

Answer ***Question No. 1*** and ***any four*** from the rest

1. Comment critically on the following statements and justify your opinion (***any four***) : 4×5

(a) Horizontal fragments are necessarily disjoint in case of a spatially distributed database.

(b) Join operation should be done after selection and projection operations and before the union operations for efficient query processing on a distributed database.

(c) Roll-up and slicing are effectively the same operations as both of these effect dimensional reduction on a data warehouse.

(d) Concept hierarchy implies a partial order between the elements.

(e) Query processing for distributed databases involves lower communication cost in general.

(f) Distributed databases offers increased availability for any database application.

2. (a) Define min-term predicate.

(b) Briefly describe the methodology for identification of primary horizontal fragments towards the design of a distributed database system.

(c) Explain the significance of derived fragmentation towards efficient design of a distributed database system, using a suitable example.

(d) In the context of distributed database design, why is it important to identify which part of application data are read-only items ? 2+6+8+4



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3. (a) Explain the role of semi-join for query optimization in a distributed environment using an example.

(b) Zeus Healthcare Society (ZHS) has a chain of hospitals in big cities across India. The information to be stored is given below :

- Hospital information: hospital id, city, number of beds, bed charges.
- Patient information: patient id, name, address, phone number
- Occupancy: patient id, hospital id, bed number, admission date, release date.

Design a distributed database solution for hospital bed reservation system to be accessed from multiple terminals in multiple wards located in different hospitals of ZHS considering the following frequent queries/operations :

- Request for reservation of bed. If a patient wants to book multiple beds in same or more than one hospital of ZHS, for different period, separate reservation is to be done for each hospital bed and period of booking.
- Availability of bed - the system is to check for availability of bed in a particular hospital for a period.
- Revenue query - the system is to be used to enquire about the revenue generated in a single hospital or in the entire chain for a period.

Your design should include the definition of global schema, fragmentation schema and allocation schema. State your assumptions, if any. 8+12

4. (a) What is distribution transparency ? How is it achieved for distributed databases in multiple levels ?

(b) State two main disadvantages of distributed databases.

(c) Define physical image of a global relation at a site.

(d) Explain the role of local mapping schema towards integration of a heterogeneous multi-site database environment.

(e) Explain with an example how a single updation operation on a single data item could result in multiple updation operations on several sites when the database is physically distributed.

(f) What is semi join ? 4+3+1+6+5+1

5. (a) Define the following terms in the context of building a data warehouse :

(i) Extraction

(ii) Transformation

(iii) Loading

(iv) Integration

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(b) Why granularity of data is an important aspect towards efficient designing of a data warehouse ?

(c) How Snow-Flake is different from a Star Schema for data warehouse ?

(d) A cube model has data in four dimensions for a particular subject orientation. The cardinalities for the four dimensions are 3, 4, 5 and 4 respectively.

(i) How many Fact and Dimension Tables are required in a equivalent Star schema ?

(ii) State and explain the minimum number of tuples that would occur in each of Dimension and Fact Tables in your design.

(e) How do you relate a lattice with a data warehouse ? 4+1+3+6+6

6. (a) What is semi-structured data ?

(b) Why newer conceptual data models are required for semi-structured data ?

(c) Akiho's Kitchen is a fast food chain selling Japanese food. Different dishes sold in AK require different number of ingredients in diverse proportion. The employee profile too varies depending on the types of employees and their job functionality.

We assume that a Hyper file named *Dishes* stores the basic information like item-name, chef, short-description and cost for different dishes. Assume that some of the entries in the Hyper file *Dishes* point at some other entries as FOT (frequently ordered together), if such dishes are frequently ordered together.

(i) Write a query using Hyper file model to list the name and salary for all the Chefs who work in Akiho's Kitchen.

(ii) Write a query using Hyper file model to extract items that are frequently ordered together with Sashimi. Your query must extract all the items that are linked up to third level considering Sashimi in level 1.

(iii) State your assumptions clearly for the additional Hyper files in the database other than *Dishes* that you need to define for answering the queries. 2+4+[4+6+4]

7. (a) What is a check-point? How is it helpful for transaction processing ?

(b) Explain the roles of distributed transaction manager (DTM) and local transaction manager (LTM) towards transaction management.

[ Turn Over ]

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(c) A "commit" operation in database systems ensure that the changes in the database are permanent. In the 2-phase commit protocol for transaction management, two different types of "commit" operations are performed at the coordinating site and at the participating sites. The coordinating site may decide to abort a transaction even if one or more participating site is ready for "local commit". Explain if this is consistent with the ACID properties for transaction management.

(d) "Isolation facilitates concurrent execution of transactions" — justify the correctness of the statement.

(e) How the atomicity of a transaction is ensured under 2-phase commit protocol when one of the sites fails after the coordinator issues a global commit message ?

(f) Compare primary copy locking versus distributed 2-phase locking. 2+4+3+5+3+3

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